

1 UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

2 DOCTORAL THESIS

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4 **The measurement of Higgs**
5 **production cross section in the**
6 **four-lepton final state at CMS**

7

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Chapter 1

Theoretical Background

In this chapter, theoretical grounds upon which this dissertation is built, are briefly discussed. Under the umbrella of “elementary particle physics”, we have been able to get answers to long lasting questions about the visible universe and even beyond. We have been able to know about the smallest structures of which the universe is made and forces which govern them to bind together in the form of ordinary matter. Standard Model of the particle physics is a theoretical framework capable to explain the dynamics of this subatomic world of particles in its beautiful mathematical formulations which are based on symmetries. Higgs boson is manifestation of arguments from core components of Standard Model which was first discovered at CERN, LHC facility in proton proton collisions. In next Sections, details on Standard Model, its Higgs sector are described. Additionally, underlying theoretical concepts on collision experiment (i.e. proton-proton) are discussed.

1.1 The Standard Model

The Standard Model (SM) of particle physics, is the theory which has provided modern concept of understanding the dynamics of the constituents of the matter and underlying principles through which they interact with each other. It is capable to give unified explanation of electromagnetic (EM), weak and strong forces (except the gravity) [1].

1.1.1 From the History

SM is a quantum field theory (QFT) belonging to $SU(3)_C \otimes SU(2)_L \otimes U(1)_\gamma$ gauge symmetry group. It was formulated from a collaborative work spanned throughout the 20th century. In 1961, there was a proposal from Glashow that electromagnetic and weak interactions are similar substructures of electroweak interaction and presented their unified interpretation with $SU(2)_L \otimes U(1)_\gamma$ symmetry group. Brout, Englert, Higgs *et al.* proposed a mechanism with which fundamental particles may acquire mass through the idea of spontaneous symmetry breaking and associated new elementary particle called Higgs boson. Salam and Weinberg then annexed this idea to the Glashow's model ($SU(2)_L \otimes U(1)_\gamma$) and completed the EW sector of SM.

In 1964, Gell-Mann and Zweig proposed the quark model independently and at the same time Nambu *et al.*, in 1965, gave the idea of color charge an important property of a quark. In 1973, formulation of strong force as a QFT (Quantum Chromodynamics, QCD), was done with $SU(3)_C$ by Gell-Mann and Fritzsh. Earlier in 1972, Veltman and Hooft proved SM to be a renormalizable theory making perturbation theory applicable for detailed calculations and resulting predictions be tested in the experiments.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi's interaction [2]. Experimental observation of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino in 1973 by Gargamelle collaboration and then the discovery of W and Z gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 [3, 4, 5, 6]. After EW was widely accepted, the Physics Nobel Prize was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role in electroweak unification.

Meanwhile, QCD part of the SM was also being developed. There were evidences of the quarks in different scattering experiment facilities in late 60's. In 1974, existence of the quarks was further verified with the discovery of J/ψ meson. Friedman *et al.* and Richter with Ting were nominated for the Nobel prizes for the discoveries of quark and charmonium respectively. Later on, discoveries of particle (a 3rd generation lepton), bottom and anti-bottom quarks were discovered in 1976, 1977 and 1995 respectively. Top quark was finally discovered at Fermilab

in 1995. All the discoveries in this era, provided a consistent test of SM predictions. Higgs boson, however, still remained unobserved until July 2012 when CMS and ATLAS collaborations of CERN announced its discovery [7, 8]. Over this extraordinary experimental success, Higgs and Englert were awarded with Physics Nobel Prize in 2013 for their theoretical predictions about this gauge boson. Despite of an incredibly successful theory of particle world, the SM does fail to fully describe the known universe.

1.2 Fundamental Particles and underlying Interactions

1.2.1 Fundamental Particles

The elementary particles¹ of SM have certain trends and can be listed in terms of a table like a periodic table of elements in Chemistry. Fundamental particles can be categorized into fermions (characterized by Fermi-Dirac statistics) and bosons (Bose-Einstein statistics). The spin-statistics theorem classifies the particles with integral spin as bosons, while particles with half-integral spin as fermions. Fermions includes all leptons, quarks and baryons (composite particles three quarks). Ordinary matter is composed of 1st generation of fermions, i.e. leptons (e.g. electron that revolves around the nucleus) and quarks (which bind to give protons and neutrons of which nucleus is composed) which are categorized in three “generations” as illustrated in Table 1.1.

Leptons, in each doublet or generation, there is a lepton and corresponding neutrino (ν) as neutral particle. Leptons only interact via electromagnetic and weak interaction, since they have no color charge. Neutrinos are electrically neutral and carry no color charges, so interact through weak force only. Charged leptons masses are listed in order of increasing mass $m_e = 0.511$ MeV, $m_\mu = 105.7$ MeV and $m_\tau = 1.78$ GeV. Charged leptons carry -1 and their antiparticle carry $+1$ electric charge. In addition, lepton number can be assigned to the particles which is “1” for leptons and “-1” for anti-leptons. Lepton number is conserved in SM processes.

In all generations, quarks carry only fractional charges. First generation includes an “up” type quark which carries electric charge $+\frac{2}{3}$ and a “down” type

¹Particle having unknown structure is called fundamental particle.

quark carrying electric charge $-\frac{1}{3}$. Quarks also carry color charge (red, green, or blue) which is responsible for the strong interaction. The types of the quarks in order of increasing mass are “up”, “down”, “strange”, “charm”, “bottom” and “top”. The whole visible world is only made of light type of quarks, “up” and “down”. The heavy quarks (second and third generation quarks) being unstable can be produced only in high energy collisions (cosmic rays, particle accelerators) and quickly decays into “up” and “down” quarks. All quarks have a property which can be regarded as color charge which is conserved in all SM processes. Each flavor of quark can have three different colors; red, green and blue (R,G,B).

Quarks interact through all three Standard Model forces, weak, strong, and electromagnetic. According to the Quantum Chromodynamics (QCD), two quarks (quark antiquark pair) or three quarks can combine as mesons ($q_1 \bar{q}_2$) or baryons ($q_1 q_2 q_3$) respectively to make color-neutral hadrons. Out of hadrons, protons and neutrons are most widely recognized making nuclei of matter which are formed of uud and udd quarks respectively.

TABLE 1.1: Summary of different Standard Model particles, and values of parameters related to them. Masses of the particles are taken from [masses] (In SM neutrino masses are taken as zero.).

Family (generation)	Name	Symbol	Mass	Charge[e]	Interactions
leptons (1 st)	electron-neutrino	ν_e	$<2 \text{ eV}$	0	weak
leptons (1 st)	electron	e	0.511 MeV	-1	weak, EM
leptons (2 nd)	muon-neutrino	ν_μ	$<0.19 \text{ MeV}$	0	weak
leptons (2 nd)	muon	μ	105.658 MeV	-1	weak, EM
leptons (3 rd)	tau-neutrino	ν_τ	$<18.2 \text{ MeV}$	0	weak
leptons (3 rd)	tau	τ	1776.82 MeV	-1	weak, EM
quarks (1 st)	up	u	2.3 MeV	$+\frac{2}{3}$	EM, weak, strong
quarks (1 st)	down	d	4.8 MeV	$-\frac{1}{3}$	EM, weak, strong
quarks (2 nd)	charm	c	1.275 GeV	$+\frac{2}{3}$	EM, weak, strong
quarks (2 nd)	strange	s	95 MeV	$-\frac{1}{3}$	EM, weak, strong
quarks (3 rd)	top	t	173.07 GeV	$+\frac{2}{3}$	EM, weak, strong
quarks (3 rd)	bottom	b	4.18 GeV	$-\frac{1}{3}$	EM, weak, strong

Carriers of fundamental forces are bosons, more details would be described in Section 1.2.2. Electromagnetic force is mediated by photon which is massless, neutral and of spin-1. On the other hand, weak force is carried by $W^\pm$ (80.4 GeV) and Z (91.2 GeV) bosons having electric charges ± 1 and 0 respectively. At last, strong forces is carried by massless, neutral spin-1 gluons. Gluon is the only gauge

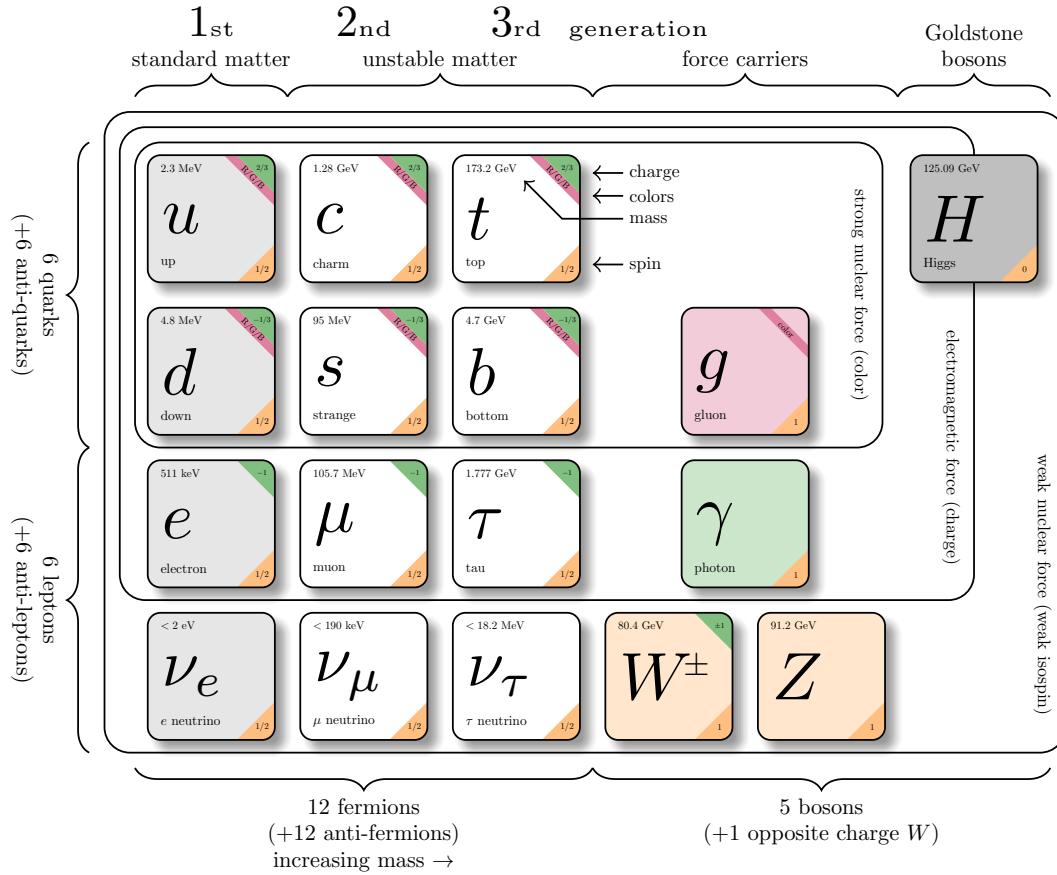


FIGURE 1.1: The elementary particles given in three generations of matter, the gauge bosons and the Higgs boson.

150 boson which carries color and anti-color charge, thus it can interact with other
 151 gluons.

TABLE 1.2: Gauge bosons and their masses, coupling constants and interaction ranges for the three forces of the SM

Force	Weak	Electromagnetic	Strong
Mediator boson	$W^\pm$ and Z	photon (γ)	Gluon (g)
Mass(GeV)	80.4, 91.2 GeV	0	0
Coupling Constant	$G_F \approx 1.2 \times 10^{-5} \text{ GeV}^{-2}$	$\alpha = \frac{1}{137}$	$\alpha_s \approx 0.1$
Interaction Range	10^{-16} cm	∞	10^{-13} cm

1.2.2 Fundamental Interactions

The three fundamental interactions (electromagnetic, strong and weak) are explained three dedicated theoretical descriptions Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Quantum Flavordynamics (QFD) which are encoded in SM.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi's interaction [2]. Weak force is mediated by $W^\pm$ and Z bosons. In terms of a field theory, it is represented by gauge theory with symmetry group $SU(2)$, having three gauge bosons (W^+, W^-, Z). Weak force, rarely known from quantum flavordynamics (QFD), is the only force which is proficient to change the flavor of quarks. This is very short range force because exchange bosons ($W^\pm, Z$) are massive. All fermions and the Higgs boson can interact via weak force. In 1970s, electromagnetic and weak forces were unified as electroweak force above unification energy (on the order of 100 GeV). The Physics Nobel Prize was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role in electroweak unification. In the field theory representation, Electroweak unification is settled under the gauge group $SU(2)_L \otimes U(1)_\gamma$. Experimental observation of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino in 1973 by Gargamelle collaboration and then the discovery of W and Z gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 [3, 4, 5, 6].

Electroweak unification

As discussed above, Glashow proved that EM and weak interactions can be expressed by $SU(2)_L \otimes U(1)_\gamma$ gauge group, where “ L ” represents left handed chiral states of fermionic fields. “ Y ” is the weak hyper charge and can be related to the weak-isospin (T) and electric charge (Q) as follows:

$$W = T_3 + \frac{Y}{2} \quad (1.1)$$

This is known as Gell-Mann-Nishijima formula and applies to left-handed and right-handed fermions.

Left handed states form weak-isospin doublets i.e. $\chi_L = (\nu_\ell, \ell)_L^T$, whereas right-handed components form weak-isospin singlet $\psi_R = \ell_R$.

To establish the local gauge invariance in the proposed theory, three vector fields (W_μ) couple to weak isospin currents with strength “g”. A neutral field denoted by B_μ couples with strength $\frac{g'}{2}$ to the weak hypercharge. To construct the Lagrangian, the covariant derivative can be written as:

$$D_\mu = \partial_\mu + ig\mathbf{T} \cdot \mathbf{W}_\mu + \frac{ig'}{2}Y B_\mu \quad (1.2)$$

This covariant derivative can be operated on the left-handed and right-handed states for the transformation as under:

$$D_\mu \chi_L \rightarrow e^{i\alpha(x) \cdot \mathbf{T} + i\beta(x)Y} D_\mu \chi_L \quad (1.3)$$

$$D_\mu \psi_R \rightarrow e^{i\beta(x)Y} D_\mu \psi_R \quad (1.4)$$

With the result of these gauge transformation, $\bar{\psi} D_\mu \psi$ remain invariant where:

$$\mathbf{W}_\mu \rightarrow \mathbf{W}_\mu - \frac{1}{g} \partial_\mu \alpha - \alpha \times \mathbf{W}_\mu \quad (1.5)$$

$$B_\mu \rightarrow B_\mu - \frac{1}{g'} \partial_\mu \beta. \quad (1.6)$$

Subsequently the Lagrangian can be constructed as :

$$\begin{aligned} \mathcal{L}_{EWK} = & \bar{\chi}_L(x) i\gamma^\mu (\partial_\mu + ig\mathbf{T} \cdot \mathbf{W}_\mu + \frac{ig'}{2}Y B_\mu) \chi_L(x) + \\ & \bar{\psi}_R(x) i\gamma^\mu (\partial_\mu + i\frac{ig'}{2}Y B_\mu) \psi_R(x) + \\ & \frac{1}{4} \mathbf{W}_{\mu\nu} \mathbf{W}^{\mu\nu} + \frac{1}{4} B_{\mu\nu} B^{\mu\nu} \end{aligned} \quad (1.7)$$

Where $\mathbf{W}_{\mu\nu}$ and $B_{\mu\nu}$ are the kinetic energy terms of gauge fields and can be written as in equations. 1.8 and 1.9 respectively.

$$\mathbf{W}_{\mu\nu} \equiv \partial_\mu \mathbf{W}_\nu - \partial_\nu \mathbf{W}_\mu - g \mathbf{W}_\mu \times \mathbf{W}_\nu \quad (1.8)$$

$$B_{\mu\nu} \equiv \partial_\mu B_\nu - \partial_\nu B_\mu \quad (1.9)$$

T being the generators of SU(2) which are related to the Pauli matrices as:

$$T^a = \frac{1}{2}\sigma^a; a = 1, 2, 3 \quad (1.10)$$

The Lagrangian in equation 1.7 describes of electroweak particle (massless lepton). In order to keep the theory invariant under the transformations as described in equations 1.5 and 1.6, mass term has been chosen to be zero i.e. the theory remains unable to describe the kinematics of the massive gauge bosons. Some years later, Salam *et al.* introduced a mechanism to include the mass term in the Lagrangian ($\mathcal{L}_{EWK}$ in equation. 1.7) keeping the theory anomaly free. More details are described in the next Section.

1.3 The Brout-Englert-Higgs Mechanism (BEH)

Glashow's proposal of unification of EM and weak forces described the kinematics of massless particle only as described in Section 1.2.2. However, gauge fields must acquire mass in order to correctly describe weak interaction and to be consistent with experimental findings. The Higgs mechanism was postulated as a part of SM, which introduces the existence of a new scalar Higgs field. This field transformation ensures the renormalizability of the theory and gauge invariance. By this formulation the concept of breaking of symmetry and origin of masses of the fermions and bosons ($W^\pm$, Z boson) are introduced [9, 10, 11, 12, 13, 14]. The Higgs mechanism introduces a doublet (under isospin) of complex scalar field

$$\phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^1 + i\phi^2 \\ \phi^3 + i\phi^4 \end{pmatrix} \quad (1.11)$$

which can be substituted in electroweak lagrangian given below

$$\mathcal{L}_{EWKB} = (D_\mu \phi)^\dagger (D_\mu \phi) + V(\phi^\dagger \phi) \quad (1.12)$$

here V is called field's self interaction or potential term, D_μ is described in Equation. 1.2. The lagrangian described in Equation 1.12 is gauge invariant under the $SU(2)_L \otimes U(1)_\gamma$ transformation. The kinetic energy term in the lagrangian is written in terms of the covariant derivatives whereas the potential only depends upon the product $\phi^\dagger \phi$. ϕ is the scalar field and can be codified in terms of quantum numbers which are isospin (I_3), hypercharge (Y) and electric charge (Q):

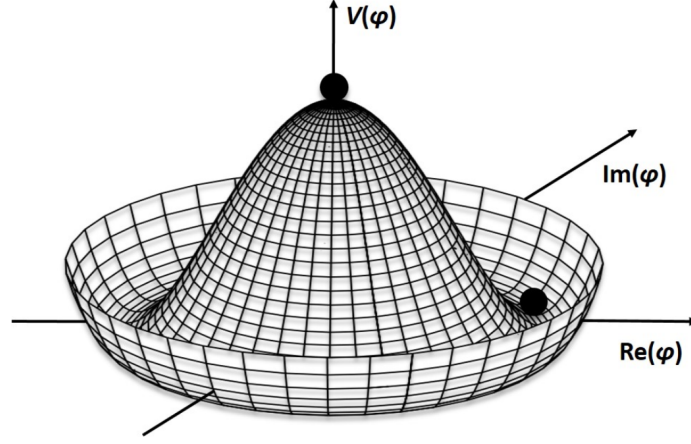


FIGURE 1.2: The Higgs potential as in Eq.1.13

220

	I_3	Y	Q
$\begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix}$	$\begin{pmatrix} \frac{1}{2} \\ \frac{1}{-2} \end{pmatrix}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

221

while the potential term in Equation 1.12 can also be expressed as

$$V(\phi^\dagger \phi) = \mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2 \quad (1.13)$$

222 here μ is mass term and the parameter λ represents the strength of the interaction.
 223 As $\mu^2 < 0$ and $\lambda > 0$ and the potential shapes becomes like a “mexican hat” very
 224 interesting, which is essential for Higgs mechanism shown in Figure 1.2. There
 225 occurs a minima for

$$\phi^\dagger \phi = \frac{1}{2}(\phi_1^\dagger + \phi_2^\dagger + \phi_3^\dagger + \phi_4^\dagger) = \frac{-\mu^2}{2\lambda} \equiv \frac{v^2}{2} \quad (1.14)$$

226 Naturally the whole system will be residing in arbitrarily chosen vacuum state;
 227 however, this state is away from the symmetry center. So, the minimum does not
 228 exist at a single value of ϕ , but at circular non-zero values. The $(\phi^\dagger \phi^0)$ selection that
 229 affiliated with the ground state is arbitrary and in $(\phi^\dagger \phi^0)$ plane the selected point is
 230 not invariant under rotations which is called spontaneous symmetry breaking. To
 231 settle the ground state on the ϕ^0 axis, the vacuum expectation value (VEV) of field

232 ϕ is

$$\langle \phi \rangle = \frac{1}{\sqrt{2}} = \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad v^2 = -\frac{\mu^2}{\lambda}. \quad (1.15)$$

233 The field ϕ given below, is derived by rewriting it in some generic gauge, on foot-
234 ings of its VEV:

$$\phi = \frac{1}{\sqrt{2}} e^{\frac{i}{v} \phi^a t_a \phi} = \begin{pmatrix} 0 \\ h(x) + v \end{pmatrix}, \quad a = 1, 2, 3 \quad (1.16)$$

here ϕ_a and $\phi_4 = h(x) + v$ represents massless fields called Goldstone fields. New degrees of freedom is introduced along with six degrees because of polarizations vector bosons $W^\pm$ and Z , as they are massless and scalar. A unitary gauge transformation in the field is fixed by

$$\phi' = e^{-\frac{i}{v} \phi^a \mathbf{T}_a} \phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ h(x) + v \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \phi^4 \end{pmatrix}$$

235 so here we have zero expectation value for the Higgs field.

236

237 The BEH Lagrangian can be written from Equation 1.12 as:

$$\begin{aligned} \mathcal{L}_{BEH} = & \frac{1}{4} g^2 v^2 W_\mu^+ W^{-\mu} - \frac{1}{2} (\partial_\mu W_\nu^+ - \partial_\nu W_\mu^+) (\partial^\mu W^{-\nu} - \partial^\nu W^{-\mu}) + \\ & \frac{1}{8} (g^2 + g'^2) v^2 Z_\mu Z^\mu - \frac{1}{4} (\partial_\mu Z_\nu - \partial_\nu Z_\mu) (\partial^\mu Z^\nu - \partial^\nu Z^\mu) - \\ & \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} \partial_\mu h \partial^\mu h - \mu^2 h^2 + \dots \end{aligned} \quad (1.17)$$

238 where W terms are written as:

$$W^\pm = \left(\frac{W^1 \mp iW^2}{\sqrt{2}} \right) \quad (1.18)$$

239 These terms correspond to the oppositely charged W bosons whose mass is
240 given by:

$$m_W = \frac{1}{2} v g \quad (1.19)$$

241 There also emerges a term for the neutral vector Z boson:

$$Z_\mu = \frac{-g' B_\mu + g W_{mu}^3}{\sqrt{g^2 + g'^2}} \quad (1.20)$$

242 Z boson mass is given by:

$$m_Z = \frac{1}{2}v\sqrt{g^2 + g'^2} \quad (1.21)$$

243 Using W_μ^3 and B_μ , we can also write their combination orthogonal to Z_μ

$$A_\mu = \frac{g'B_\mu + gW_\mu^3}{\sqrt{g^2 + g'^2}} \quad (1.22)$$

244 The Equation 1.22 corresponds to photon with the condition:

$$\frac{g'g}{\sqrt{g^2 + g'^2}} = e \quad (1.23)$$

245 From equations 1.19 and 1.21, we can have,

$$\frac{m_W}{m_Z} = \frac{g}{\sqrt{g^2 + g'^2}} \quad (1.24)$$

246 Suppose, the θ_W be weak mixing angle with definition,

$$g' = g \tan \theta_W \quad (1.25)$$

247 Using Equation 1.25, we can write the Equations 1.20 and 1.22 as:

$$Z_\mu = -B_\mu \sin \theta_W + W_\mu^3 \cos \theta_W \quad (1.26)$$

$$A_\mu = B_\mu \sin \theta_W + W_\mu^3 \cos \theta_W \quad (1.27)$$

248 With spontaneous symmetry breaking (SSB), gauge bosons (W,Z) acquire masses
 249 as a result of interaction with BEH (or Higgs) field. Also, after SSB, BEH (or Higgs)
 250 fields emerges with a physical boson particle. Lagrangian (potential term) can be
 251 written as:

$$\mathcal{L}_V = \frac{1}{4}\lambda(v + h(x))^4 - \frac{1}{2}\lambda v^2(v + h(x))^2 \quad (1.28)$$

252 This gives the Higgs mass at first order; $M_H^2 = 2\lambda v^2$

253 The value of v (also g) can be measured using Fermi weak constant i.e. G_F . How-
 254 ever, λ is a free parameter in SM.

255 Fermions can also acquire masses if additional gauge invariant interaction

term is added to term EW Lagrangian. In this way, fermions can acquire masses when they interact with BEH (or Higgs) field through Yukawa couplings. Interaction term in the Lagrangian can be written as:

$$\mathcal{L}_{Yukawa} = -G_l[\bar{\psi}_R(\phi^\dagger\chi_L) + (\bar{\chi}_L\phi)\psi_R] \quad (1.29)$$

Using ϕ and other inputs can further gives us the masses (m_f) of fermions and couplings (g_{Hf}):

$$m_f = \frac{G_l}{\sqrt{2}}v \quad (1.30)$$

$$g_{Hf} = \sqrt{G_l}\sqrt{2} = \frac{m_f}{v} \quad (1.31)$$

1.4 Phenomenology of proton-proton collisions at LHC

LHC is the largest colliding experiment of its kind ever built also motivated to perform test of the SM of particle physics. More details on LHC experiment will be discussed in next chapter.

In this section, however, some phenomenological part of proton-proton collision in high energy experiments, like LHC, are briefly discussed. Results of scattering experiments e.g. production cross section or a certain process, are compared with prediction of perturbative QCD. It is possible through hard scattering experiments to probe at very small scale even shorter than a proton radius. Hence it helps to know the interactions of quarks and gluons (collectively called as partons) processes probe distance scale. A typical diagram of the hard scattering process in proton proton collisions is shown in the Figure. 1.3 and also in Ref. [scattering]. Protons are composite particles made up of three valence quarks (uud) which are bound together with strong force exchanging the gluons. There are several ways through which they can interact within a proton. Quarks can mutually interact with other quarks or through annihilation process they can produce gluons which can further go for pair-production of quark anti-quark pair. As a result, apart from the valence quarks, there is a sea of virtual quarks and gluons inside a proton (collectively called partons). In a collision experiment, it may happen that there is large momentum transfer from the colliding partons. This is called hard scattering and as a result many resonances can be produced e.g. Higgs. Probability of a certain

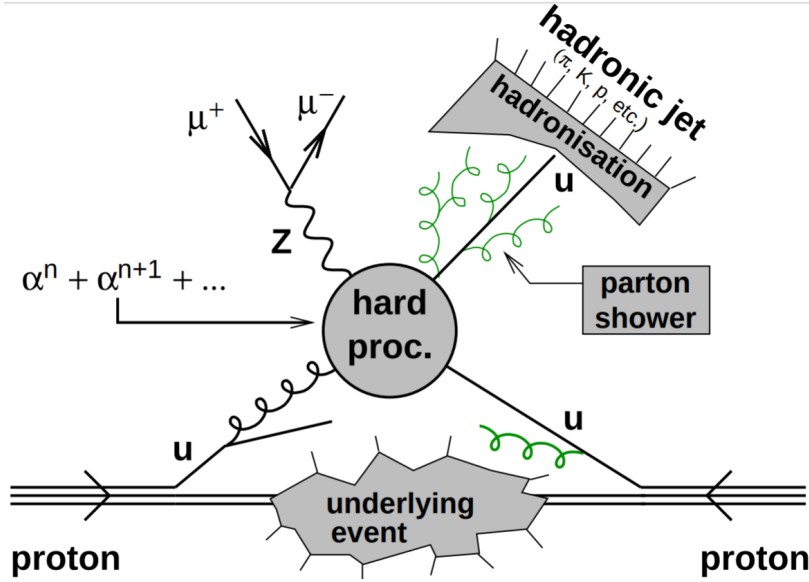


FIGURE 1.3: A schematic diagram of proton-proton collision at high energy experiments

process to occur is known as cross section of that process. The production cross sections of different process at LHC in the proton-proton collisions are shown in the Figure. ?? in next chapter.

Partons which do not take part in hard interaction, usually go straight and may involve in certain low momentum transfer processes. These activities of low momentum transfer are called underlying events. In case if more than one interactions happen in a certain proton proton collision, it is referred as multi-parton interactions (MPI).

In an accelerated environment, charged partons may radiate photons or gluons through a process known as “Bremsstrahlung”. If this radiation is associated with the incoming partons, it is called initial state radiation (ISR) and if it is coming from outgoing parton from the interaction point, it is called final state radiation (FSR). The quarks and/or gluons when they go through the process of hadronization (bind together to form color singlet) make a bunch of hadrons usually in the form of a cone which are called “jets” and are described in Section. ??.

In hard scattering experiments, probability of a certain process to occur is known as cross section and can be calculated using the Factorization theorem. This theorem helps to disintegrate the cross section into hard and soft processes using the scale which is known as factorization scale (μ_F).

Suppose in collision of two protons, A and B, the partons a and b undergo hard

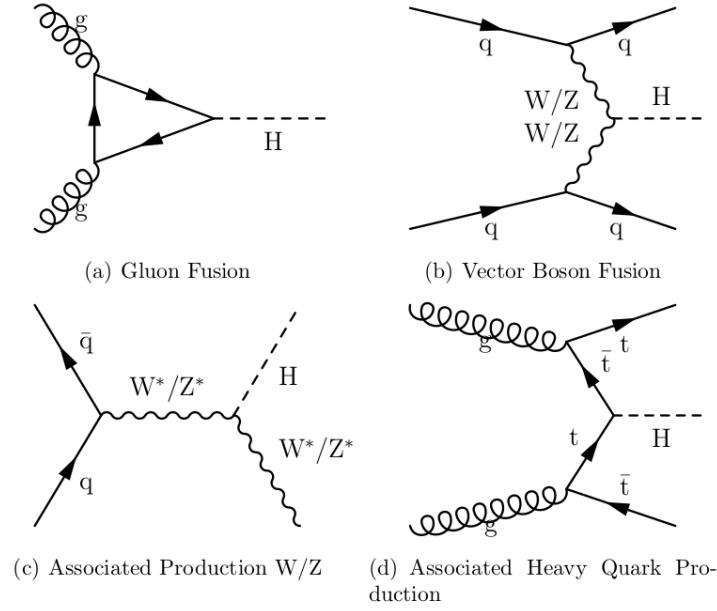


FIGURE 1.4: Feynman diagrams of Higgs production at tree level.

scattering which produce a certain final state S . The total cross section can be given by the relation:

$$\sigma_{AB} = \int dx_a dx_b f_{a/A}(x_a, \mu_F^2) f_{b/B}(x_b, \mu_F^2) \hat{\sigma}_{ab \rightarrow S} \quad (1.32)$$

In above equation, x_a is the fraction of momentum of the parton a which it takes from proton A. Similarly, x_b is the fraction of momentum of the parton b which it takes from proton B. μ_F is factorization scale and is usually of the order of Q which is total momentum transfer of hard process. $f_{a/A}(x_a, \mu_F)$ and $f_{b/B}(x_b, \mu_F)$ are the parton distribution functions (PDFs).

1.5 Higgs boson Production at LHC

Higgs can be produced at LHC in various production modes. The production modes of Higgs which are dominant in proton proton collisions at LHC are illustrated in Figure 1.4, corresponding production cross sections as a function of its mass at center-of-mass energies as shown in Figure 1.5. A considerable increase in production cross section of Higgs is found as we move towards higher center-of-mass energies [15]. The most dominant production mode of Higgs at LHC is

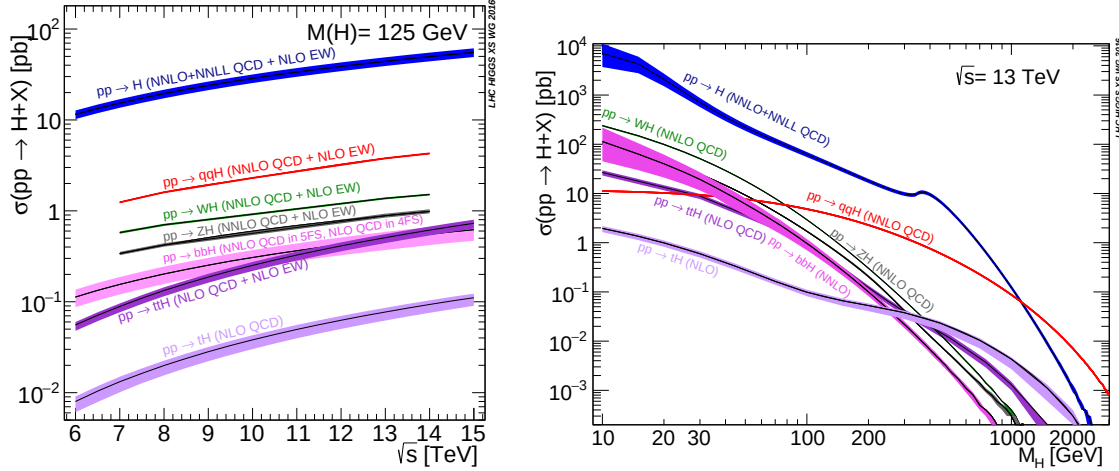


FIGURE 1.5: Production cross sections of Higgs at 13 TeV as a function of center of mass energies (left) and as a function of its mass (right).

gluon-gluon fusion as illustrated in Figure 1.4 (top left) [15]. The gluons couples to Higgs through the loop of virtual quarks, as coupling is directly proportional to fermion mass, top quark loop is most dominant. VBF is second most important production mode of Higgs at LHC and LEP as shown in Figure 1.4 (upper right). Its cross section is roughly 10 times lower than ggH at $m_H = 125$ GeV, and becomes comparable at $m_H = 1$ TeV. In VBF production mode, two $W^\pm$ or Z boson fusion create Higgs that have been emitted by a (anti)fermion pair collision. The VBF experimental signal is very clean and can be well distinguished from ggH with two high energy jets in detected in forward region. These jets signature helps to increase signal to background ratio.

In W/Z association production mode, Higgs is radiated by high energetic $W^\pm$ or Z bosons that is produced by fusion of two fermions. This production mode was leading mechanism at LEP and sub-leading mechanism at Tevatron. At LEP it becomes very probable to produce $W^\pm$ or Z boson with electron positron collisions. In case of hadron collider, Tevatron makes quark antiquark collisions more likely as compared to LHC. The least dominant process of Higgs production is illustrated in Figure 1.4 (bottom right). In this process two gluons collide producing pair of top antitop which produces Higgs by fusion in association $t\bar{t}$.

1.6 Higgs Decay

Higgs, being an electroweak particle, can decay in variety of channels depending upon mass of final state particles, interaction strength and etc. The life time of SM 126 GeV Higgs is 1.6×10^{-22} s and it can decay to all SM particles having masses (most probably to heaviest onshell particle). Higgs boson decay branching ratio to a variety of particles are illustrated in Figure 1.6 [15]. $H \rightarrow b\bar{b}$ (at $m_H \approx 125$ GeV) is the most dominant decay mode of Higgs. But the QCD background in this channel is of many order higher than signal which makes it less sensitive channel. However, CMS and ATLAS collaborations and other collaborations [16], [17] and [18](CDF and D0) have exploited this channel, in association with a vector boson which decays to leptons, in boosted regime. B quarks are also difficult to reconstruct in a detector. When they decay in the detector, they fragment into jets of many hadrons. These jets are difficult to reconstruct perfectly. Because of this, it is necessary to exploit other channels in this region. $H \rightarrow \gamma\gamma$ is one most favorable decay channel due to signature with two high energy photons with excellent mass resolution.

As the Higgs mass at 125 GeV, Higgs boson decay to diboson(WW , ZZ) boson are also available, where one or both boson can be off-shell. WW decay further has neutrinos in final state due to which it is not possible to reconstruct the final state fully. $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ decay is labeled as golden channel in spite of low branching ratio. The clean signature of at least four leptons makes this channel almost background free. These leptons are high energetic leptons which can be reconstructed with excellence in the detector [15]. Measurements on Higgs reported in this dissertation are based on $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$ channel.

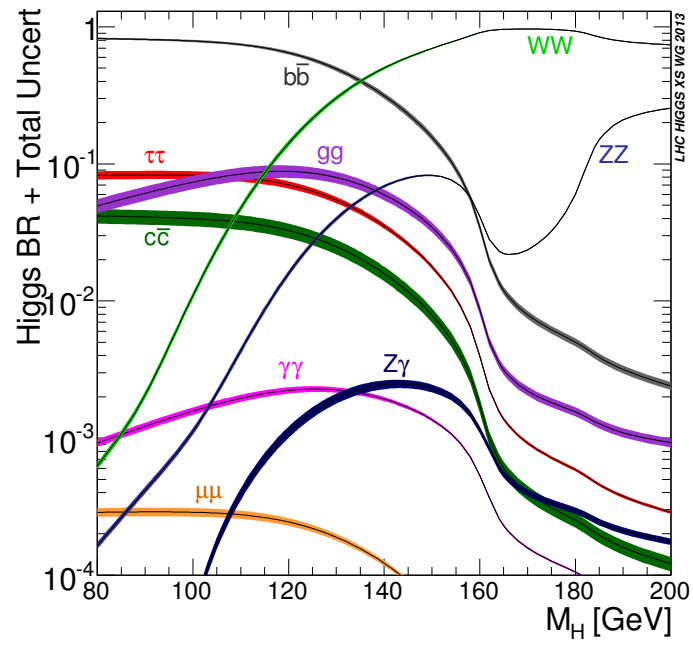


FIGURE 1.6: Higgs branching ratios for different channels (in terms of mass)

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